

Proof 2

Review

1. $y' = -2x + 5$

At the turning point, $y' = 0$. Then

$$-2x + 5 = 0$$

$$2x = 5$$

$$x = \frac{5}{2}$$

$$\begin{aligned}\therefore y &= -\left(\frac{5}{2}\right)^2 + 5\left(\frac{5}{2}\right) + 2 \\ &= 8.25\end{aligned}$$

So the coordinate is $\left(\frac{5}{2}, \frac{33}{4}\right)$.

To find whether the vertex is a minimum or a maximum:

$$y'' = -2$$

Since the second derivative is less than zero, the vertex is a maximum.

2. $\sqrt{36}$ (D)

3. By inspection:

$$2x^3 + x^2 - 5x + 2 = (x - 1)(2x^2 + 3x - 2)$$

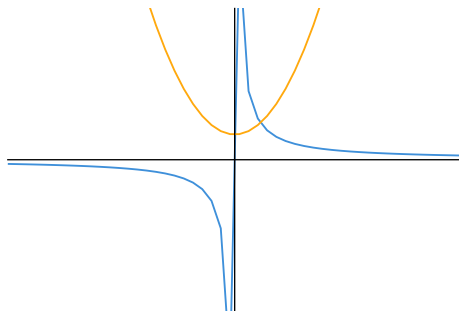
So $a = 2$, $b = 3$, $c = -2$

4.
$$\begin{aligned}(1 - 2x)^{12} &\approx 1 + 12(1)(-2x) + 66(1)(-2x)^2 + 220(1)(-2x)^3 \\ &\approx 1 - 24x + 264x^2 - 1760x^3\end{aligned}$$

When $x = 0.01$:

$$\begin{aligned}(0.98)^{12} &\approx 1 - 24(0.01) + 264(0.01)^2 - 1760(0.01)^3 \\ &\approx 0.7846\end{aligned}$$

5.



Therefore there is only one root to the equation.

Consolidation

1. Claim: if n is even, $n^2 + n$ is even. If n is even, n^2 is even because an even number multiplied by an even number is even. Then $n^2 + n$ is even because an even number added to another even number is even.

$$\begin{aligned}n &= 2k \\(2k)^2 + 2k &= 4k^2 + 2k \\&= 2(2k^2 + k)\end{aligned}$$

Therefore if n is even, $n^2 + n$ is even.

2. i. Let k be an integer. Then $k + 2k + 3k + 4k + 5k = 15k = 5(3k)$, therefore the sum of five consecutive integers is a multiple of five.
ii. The mean of five consecutive integers is $\frac{k+2k+3k+4k+5k}{5} = 15\frac{k}{5} = 3k$ which is the third number.

3. i. Claim: the difference between any two square numbers is odd.

But $n^2 - n^2 = 0$ which is even, so this claim is not true.

- ii. Claim: all prime numbers are odd.

But 2 is a number greater than 1 that has factors of only 1 and itself, so the claim is false.

- iii. Claim: for any positive integer n , the number $n!$ has 2^{n-1} distinct factors.

But $6!$ has 30 factors: the prime factorization of 720 is $2^4 \times 3^2 \times 5^1$ so the number of factors it has is $(4 + 1)(2 + 1)(1 + 1) = 30$, but 2^5 is 32. Therefore, the claim is false.

4. Months with 5 or more characters:

- i. January: jury
- ii. February: bear
- iii. March: arch
- iv. April: fair
- v. August: guts
- vi. September: bent
- vii. October: boot
- viii. November: even
- ix. December: code

5. Let $100 + n$ be a number greater than or equal to 100 where $n \geq 0$. Then for the number to be divisible by 4, n must also be divisible by 4 because $4(25 + \frac{n}{4})$ must be a whole number for the number to be divisible by 4. Therefore, n is divisible by 4.

6. 27 is not prime and its prime factors are 3^3 , so it is not prime and the product of two primes.

Exam Questions

1. Claim: "If $n \geq 0$ and $n \in \mathbb{Z}$, $3^n + 2$ is prime."

But when $n = 5$, $3^n + 2 = 245$ and $5 \mid 245$ so the claim is incorrect because 245 is not prime.

2. Let $2n, 2n + 1$ be two consecutive integers. Then the sum of the two squares is $4n^2 + (2n + 1)^2 = 8n^2 + 4n + 1 = 4(2n^2 + n) + 1$ which takes the form $4k + 1$.

3. Claim: If the sum of the digits of a 2-digit number is 5, then this 2-digit number is not a perfect square.

$$5 + 0$$

50 is not a perfect square.

$$4 + 1$$

41 and 14 are not perfect squares.

$$3 + 2$$

32 and 23 are not perfect squares.

Therefore the statement is true.

4. a. Claim: for $n : n \in \mathbb{Z}$ and $20 \leq n \leq 40$, when the digits of n are squared and summed, the sum is an odd number.

But when $n = 31$, the sum of the digits squared is $3^2 + 1^2 = 10$ which is even. Therefore the claim is false.

- b. Let $n, (n + 1), (n + 2)$ where $n \geq 0$ and $n \in \mathbb{Z}$ be three consecutive positive integers. Then the sum of these integers is $n^2 + (n + 1)^2 + (n + 2)^2 = 3n^2 + 6n + 5$.

$3n^2 + 6n = 3(n^2 + 2n)$ which is divisible by 3. By adding 5 to this number, the number is not divisible by 3 because 5 is not divisible by 3.